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DEPARTMENT OF DEFENSE
MANUFACTURING PROCESS STANDARD

CONTROL OF HARDWARE CLEANLINESS

AMSC N/A

FSC 20GP

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FOREWORD

1. This Manufacturing Process Standard is approved for use by the Naval Sea Systems Command (NAVSEA), Department of the Navy, and is available for use within the distribution limitations noted on the cover page.

2. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navy.mil with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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CONTENTS

PARAGRAPH **PAGE**

1. SCOPE 1
1.1 Purpose 1
1.2 Applicability 1

2. APPLICABLE DOCUMENTS 1
2.1 General 1
2.2 Government documents 1
2.3 Non-Government publications 2
2.4 Order of precedence 3

3. DEFINITIONS 3
3.1 Approval, Approved 3
3.2 Authorized Representative 3
3.3 Cleanliness boundary 3
3.4 Cleanliness cap, cover, plug or seal 3
3.5 Corrosion resistant material 3
3.6 Crevice 3
3.7 Critical surface 3
3.8 Exposed location 3
3.9 Final cleaned surface 3
3.10 Final closure 4
3.11 Final surface 4
3.12 Foreign material 4
3.13 Hardware 4
3.14 Inaccessible area 4
3.15 Internal surface 4
3.16 Lint-free cloth 4
3.17 Loss of cleanliness 4
3.18 Loss of cleanliness control 4
3.19 Mechanical cleaning 4
3.20 Noncorrosion resistant material 4
3.21 Organization 4
3.22 Rust 4
3.23 Support system 5
3.24 Thermal treatment 5
3.25 Visual inspection 5

4. GENERAL REQUIREMENTS 5
4.1 Conflicts 5
4.2 Procedures 5

5. CONTROL OF DETRIMENTAL MATERIALS 5
5.1 Control of detrimental materials 5

6. CLEANING REQUIREMENTS 5
6.1 Cleaning 5
6.2 Cleaning restrictions on specific types of hardware 5
6.3 Cleaning methods 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:36Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH **PAGE**

6.4 Rust removal 8
6.5 Tape residue removal 8
6.6 Drying 8

7. CLEANLINESS INSPECTION 9
7.1 Introduction 9
7.2 Cleanliness inspection requirements 9
7.3 Cleanliness inspection methods and acceptance criteria 11

8. MAINTAINING CLEANLINESS 15
8.1 Introduction 15
8.2 Sealing and capping hardware openings 15
8.3 Cleanliness identity 15
8.4 Tool and equipment cleanliness 15
8.5 Preventing foreign material from entering hardware 15
8.6 Tape 18
8.7 Water requirements 19
8.8 Support system requirements 21

9. RECOVERY FROM A LOSS OF CLEANLINESS OR LOSS OF CLEANLINESS CONTROL 24
9.1 Procedure 24
9.2 Inspection for foreign material 24
9.3 Removal of foreign material 24
9.4 Verification of cleanliness 25
9.5 Records 26
9.6 Hardware acceptance 26

10. NOTES 27
10.1 Changes from previous issue 27

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Check the source to verify that this is the current version before use.

CONTENTS

FIGURES **PAGE**

1 Nonexpandable caps and seals 28
2 Nonexpandable plugs 29
3 Expandable plug 30
D-1 Crevice fixture 48

TABLES

I Final cleaning solvents 7
II Solvent temperatures for vapor degreasing 8
III Conditions determining need for independent inspection 10
IV Water purity 20
V Required grade of test water 21
A-I Acid cleaning methods 35
A-II IGA depth limits 37

APPENDICES

A CLEANING METHODS 31
B FREE IRON DETECTION 40
C REQUIREMENTS FOR CAPS, COVERS, SEALS AND PLUGS 43
D QUALIFICATION OF ALTERNATE FINAL CLEANING SOLVENTS 46

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1. SCOPE

1.1 Purpose. The purpose of this standard is to establish uniform requirements for cleaning and maintaining cleanliness of hardware. Equipment supplied under this standard shall be suitably clean for installation directly into a fluid system without further cleaning. It is extremely important that contamination be kept to a minimum to prevent adverse effects upon the system such as accelerating the corrosion and wear of equipment or changing the chemistry of the fluid in the system.

1.2 Applicability. This standard, when invoked by contract technical requirements, applies only to hardware internal surfaces. The detrimental material control requirements of MIL-STD-2041 apply to hardware internal surfaces and external surfaces of pressure boundary parts.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in Sections 3 through 9 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in Sections 3 through 9 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

TT-I-735	-	Isopropyl Alcohol
MIL-DTL-24671	-	Cloth, Lint-Free, Flushing and Cleaning

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2041	-	Control of Detrimental Materials

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

NAVSEA 250-1500-1	-	Welding Standard
NAVSEA S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessel Welds

(Copies of other Government publications should be obtained from the contracting activity or as directed by the contracting officer.)

CODE OF FEDERAL REGULATIONS (CFR)

27 CFR Part 21 - Formulas for Denatured Alcohol and Rum

(Copies of this document are available online at www.ecfr.gov.)

OTHER

AAFCSL - Approved Alternate Final Cleaning Solvents List

(Copies of this document are available online at https://logistics.unnpp.gov/ConsolidatedAPLQPLjsprod.asp.)

2.3 Non-Government publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN CHEMICAL SOCIETY (ACS) SPECIFICATIONS

Reagent Chemicals

(Copies of this document are available online at www.acs.org.)

ASTM INTERNATIONAL (ASTM)

ASTM D329	-	Standard Specification for Acetone
ASTM D4080	-	Standard Specification for Trichloroethylene, Technical and Vapor-Degreasing Grade
ASTM D4376	-	Standard Specification for Vapor-Degreasing Grade Perchloroethylene

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL (SAE)

SAE AS8660	-	Silicone Compound NATO Code Number S-736

(Copies of this document are available online at www.sae.org.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Approval or approved. Unless specifically stated otherwise, approval or approved means action taken by the Naval Sea Systems Command (NAVSEA) or its authorized representative.

3.2 Authorized representative.

(a) The Commander of a naval shipyard, for vendors to the shipyard.
(b) NAVSEA, for commercial and naval shipyards.
(c) The shipbuilder, for commercial shipyard subcontracted acquisitions.
(d) That Prime Contractor of NAVSEA, other than a shipbuilder, who has design or procurement responsibility, for vendors to the Prime Contractor.

3.3 Cleanliness boundary. For support systems, a filter arrangement or other device (such as a fluid separator) used to separate a portion of the system that is a lower level of cleanliness from a portion of the system with a higher level of cleanliness.

3.4 Cleanliness cap, cover, plug, or seal. For open or exposed hardware, temporary equipment or materials that are applied to the hardware to exclude foreign material.

3.5 Corrosion resistant material. A metal or alloy that inherently resists oxidation or chemical attack in air and water at ambient temperatures. Examples include chromium-nickel corrosion resistant steel, nickel-base, cobalt-base, and copper-base alloys, and precipitation hardened corrosion resistant steel.

3.6 Crevice. Any narrow opening in a surface or any open juncture between mating surfaces in which detrimental solutions or foreign particles can likely be entrapped and not readily observed or removed. Examples include the annular space in threaded connections and socket assemblies, tube-to-tube support joints, and small holes and grooves. Tightly fitting parts with no visible gap (such as a shrink fit or press fit) need not be considered a crevice with respect to foreign particles, but are still considered to be a crevice with respect to liquids.

3.7 Critical surface. A surface that, when corroded or otherwise roughened, can impair the hardware's proper function. Examples include small orifices, bearings, bellows, seating surfaces, seal weld diaphragm closure parts, thread surfaces, seal membranes, or heat exchanger tubing.

3.8 Exposed location. Any opening in hardware that exposes an unprotected internal crevice or inaccessible area. Examples of hardware that have an exposed location include a pump casing with is suction adapter installed and a valve body with its seat ring installed.

3.9 Final cleaned surface. A final surface of hardware on which all planned cleaning operations have been performed.

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3.10 Final closure. Any final assembly operation that will prevent visual access to internal surfaces, except that assembly operations that create crevices only are not considered final closures. Examples of final closure include installation of a lift check valve disc into the valve body and final fitup of piping joints for welding.

3.11 Final surface. A hardware surface that will be subjected to no further material removal or additive operations.

3.12 Foreign material. Any material or object that should not be on or within clean hardware; includes all detrimental material. Examples include grit, chips, particles, oil, machining coolant, liquid penetrant, dried liquid residue/stains (except water spots known to result solely from Grade B or better water and are inaccessible for cleaning or where their removal would likely damage the hardware), slag, scale, fibers, tape, tools, and loose articles.

3.13 Hardware. Any article, piece of material, individual part, subassembly, assembly, equipment, or system to which this standard applies.

3.14 Inaccessible area. A hardware surface that is not readily accessible for cleaning or visual inspection, wipe cloth inspection, or swab inspection; includes all crevices.

3.15 Internal surface. A surface that contacts system fluid.

3.16 Lint-free cloth. A cloth that is free from loose thread, fluff from cloth or yarn, or other particles of a size visible to the unaided eye, and that does not leave any visible residue on the hardware surface. If used for final cleaning, inspecting a hardware surface, or free iron checks, it shall meet the detrimental material requirements of MIL-STD-2041. Cloths need not meet the detrimental material requirements of MIL-STD-2041 if they are only being used for preliminary cleaning.

3.17 Loss of cleanliness. A condition in which a final cleaned surface of hardware does not meet the cleanliness verification criteria of this standard.

3.18 Loss of cleanliness control. A condition in which control of proper cleanliness maintenance for a final cleaned surface of hardware is lost. Loss of cleanliness control may or may not involve an actual loss of cleanliness.

3.19 Mechanical cleaning. Cleaning methods that may remove some amount of hardware material along with the foreign material being removed. Examples of mechanical cleaning include grinding, buffing, polishing, wire brushing, sanding, filing, and rubbing.

3.20 Noncorrosion resistant material. A material that inherently does not resist oxidation or chemical attack in air or water at ambient temperatures. Examples include carbon steels, low alloy steels, and low chromium non-austenitic steel (generally AISI 400 and 500 series steels, and their cast equivalents, except for AISI 403, 410, and CA6NM, which are corrosion resistant).

3.21 Organization. Any company or activity performing work within the scope of this standard.

3.22 Rust. Visible corrosion product consisting of hydrated oxides of iron. Rust usually forms when iron or ferrous alloys are exposed to water, or moisture and oxygen in air.

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3.23 Support system. A structure of fluid-handling or gas-handling equipment and/or instrumentation that is temporarily attached to the hardware that is required to be clean to this standard, that could add or recirculate fluid or gas to the hardware for the purpose of cleaning, calibrating, flushing, testing, drying, or performance of other specified operations.

3.24 Thermal treatment. Any operation, except vapor degreasing in perchloroethylene, where the temperature of the hardware exceeds 350°F (177°C). Examples include heat treating, welding, brazing, preheating for welding or brazing, hot forming, stress relieving, hot tests, and baking for embrittlement relief. Metal removal (other than thermal cutting) is not a thermal treatment.

3.25 Visual inspection. An inspection performed by the unaided eye (1X unless otherwise specified) with normal visual acuity (natural or corrected) while using at least normal reading-level lighting. Boroscopes, videoprobes, mirrors, and other aids of sufficient resolution may be used to extend the ranges of visual inspection; however, foreign material only visible at greater than 1X magnification does not violate the requirements of this standard.

4. GENERAL REQUIREMENTS

4.1 Conflicts. Any conflict noted within this standard or between this standard and other contract technical requirements shall be promptly brought to the attention of NAVSEA or its authorized representative, prior to any action by an organization.

4.2 Procedures. Each organization shall develop procedures for hardware cleaning and cleanliness control that satisfy the requirements of this standard. The procedures shall be available for review or approval upon request.

5. CONTROL OF DETRIMENTAL MATERIALS

5.1 Control of detrimental materials. The detrimental material control requirements of MIL-STD-2041 apply.

6. CLEANING REQUIREMENTS

6.1 Cleaning. Hardware shall be final cleaned by one or more of the cleaning methods in 6.3, to conform to the acceptance criteria of Section 7. Other solvent cleaning methods used in accordance with the restrictions of MIL-STD-2041 may be used for preliminary cleaning of hardware, provided that final cleaning is completed at a later stage of manufacturing and is in accordance with 6.2.1. The cleaning method selected should be effective for removing the particular foreign material present.

6.2 Cleaning restrictions on specific types of hardware.

6.2.1 Hardware surface with a crevice or inaccessible area. Each hardware surface that will be part of a crevice or inaccessible area shall be in a final cleaned condition prior to the assembly operation that creates the crevice or inaccessible area. Once assembled, cleaning of the crevice or inaccessible area requires disassembly or use of an approved cleaning procedure. Hardware shall not be manufactured with the expectation of cleaning only after completion of fabrication or assembly, except when the finished hardware is completely accessible for cleaning and inspection (that is, free of crevices and inaccessible areas). Post

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assembly flushing is not an acceptable alternative to cleaning and maintaining hardware cleanliness up to and including assembly, unless approved per Section 9.

6.2.2 Non-metallics. Non-metallic hardware shall not contact any solvent that would degrade it.

6.2.3 Acid cleaning, blast cleaning, and tumbling. Acid cleaning, blast cleaning, and tumbling shall be used only after obtaining approval. When approved for use, acid cleaning, blast cleaning, and tumbling shall be performed in accordance with the procedure requirements of A.5, A.6, and A.7, respectively.

6.2.4 Seal membranes. The final surfaces of seal membranes shall not be acid cleaned and shall not be mechanically cleaned by power driven grinding, power driven polishing, or power driven wire brushing.

6.2.5 Tube and pipe. Internal surfaces of tube or pipe that contain no crevices may be final cleaned by swabbing as specified in A.4.

6.3 Cleaning methods.

6.3.1 Mechanical cleaning. Mechanical cleaning shall be performed as described in 3.19. Mechanical cleaning shall be performed in such a manner that particles will not enter an exposed location of the hardware being cleaned.

6.3.2 Solvent cleaning. Only solvents meeting the specifications and additional requirements listed in Table I shall be used for final cleaning. Except as specified in 6.2, cleaning shall be performed with any of the solvents listed in Table I by immersion, vapor degreasing (as applicable), ultrasonic, or by wiping with a saturated clean, lint-free cloth or swab. After solvent cleaning, hardware shall be thoroughly dried.

6.3.2.1 Solvent cleaning exception. Specific solvents listed on the AAFCSL may also be used for final cleaning hardware in accordance with the restrictions specified (that is, for use on all hardware or hardware with no crevices or inaccessible areas). Activities wishing to use a specific solvent not on the list should qualify the solvent to the procedure in Appendix D and request approval prior to use.

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TABLE I. Cleaning solvents

Code	Solvent	Specification	Additional requirements
A	Acetone	ASTM D329	Unused or redistilled
B	Denatured alcohol	27-CFR-21	Denatured ethanol conforming to formula 3-A, 3-C, 23-A, or 30
C	Isopropyl alcohol	TT-I-735	Grade A or B
D	Grade B or better water	Table IV (see 8.7.1)	For loose particles or water-soluble contaminants
E	Perchloroethylene (Tetrachloroethylene)	ASTM D4376	Shall not be used on hardware with crevices or inaccessible areas or non-metallic hardware.
F	Trichloroethylene	ASTM D4080	Shall not be used on hardware with crevices or inaccessible areas or non-metallic hardware.
G	Nonionic detergent	MIL-D-16791	Maximum of 1 fluid ounce (28.3 grams [29.6 ml]) per gallon (3.8 liters) of grade C water. Shall not be used on hardware with crevices or inaccessible areas. Cleaning shall be per 6.3.2.3.
H	Trisodium phosphate (TSP) - detergent	O-S-642	Add 2.3 to 23 ounces (65.2-652 grams) of dodecahydrate TSP solution to 1 gallon (3.8 liters) of solvent H. Shall not be used on hardware with crevices or inaccessible areas. Cleaning shall be per 6.3.2.3.

Note: American Chemical Society reagent grade chemicals, if available for the solvents listed above, may also be used.

6.3.2.2 Vapor degreasing. Vapor degreasing shall be performed as follows:

(a) Solvent operating temperatures for vapor degreasing shall be maintained in accordance with Table II requirements. A solvent shall be replaced if its boiling point exceeds the applicable maximum operating temperature specified in Table II or if it is not degreasing properly.

(b) Solvents E and F shall contain a neutral inhibitor to maintain a pH of 6.5 to 7.5.

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TABLE II. Solvent temperatures for vapor degreasing.

Solvent Code	Operating Temperature, °F (°C)	
	Minimum	Maximum
E	248 (120)	258 (126)
F	185 (85)	195 (91)

6.3.2.3 Trisodium phosphate and detergent cleaning. Trisodium phosphate or detergent cleaning shall be performed as follows:

a. Hardware may be cleaned by scrubbing with or immersion in solvents G or H. A wetting agent may be added to the solution if necessary. The wetting agent, when diluted to the concentration in the detergent solution, shall conform to the detrimental material requirements of MIL-STD-2041.

b. Hardware shall be final rinsed with Grade B water at a temperature of 120°F (66°C) minimum, until no visible solvent G or H film remains on the surface after drying. Grade C water may be used for an initial rinse, if desired. Final rinsed hardware shall be dried.

c. Solutions of solvent G or H in continuous or intermittent use (e.g., cleaning baths) shall be tested weekly for chloride buildup. Sampling may be suspended during extended periods of non-use but shall be reinitiated prior to putting the solution back in use. A batch of solution shall not be used if its chloride ion concentration exceeds 500 ppm.

6.3.3 Vacuum cleaning. Hardware surfaces contaminated with dirt, grit, dust, or similar material may be final cleaned by vacuuming.

6.3.4 Flushing. Flushing to establish cleanliness or to remove foreign material to reestablish cleanliness is allowed when the criteria of 9.3.b. have been met. Flushing shall be performed in accordance with the requirements in 7.3.4.

6.4 Rust removal. Rust shall be removed using any mechanical methods of this standard. Rust may also be removed using chemical methods in accordance with MIL-STD-2041. For carbon and low alloy steels, the stripping procedure of A.3, may be used, observing the hardware restrictions for acid cleaning of A.5.

6.5 Tape residue removal. Tape residue shall be removed with any cleaning method allowed in Section 6. In addition, tape residue may be removed with the procedure of A.8 provided that the n-heptane or degreasing solvent is used in accordance with MIL-STD-2041.

6.6 Drying. When drying of hardware is specified, open flames shall not be used and hardware shall not be exposed to damaging temperatures. Drying systems should meet the support system cleanliness requirements of 8.9.

6.6.1 Special drying requirements for noncorrosion resistant material. Hardware that contains noncorrosion resistant material shall be dried as soon as practicable after water contacts the noncorrosion resistant material. Accessible surfaces of dried hardware containing noncorrosion resistant material shall be visually inspected for residual water and rust.

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7. CLEANLINESS INSPECTION

7.1 Introduction. This section provides cleanliness inspection requirements, methods, and acceptance criteria to determine that the cleanliness requirements of this standard have been met. Hardware without inaccessible areas that is cleaned to MIL-STD-2041 can be accepted as clean to this standard.

7.2 Cleanliness inspection requirements.

7.2.1 Inspection requirements. Hardware surfaces shall be inspected for cleanliness at the times and by the personnel specified in Table III.

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TABLE III. Conditions determining need for independent inspection.

CODE	INSPECTION	INDEPENDENT INSPECTION REQUIRED
a.	Following entry into clean hardware.	Yes, for entry that is dirt producing (such as machining or grinding) or potentially dirt producing operations (such as clean cuts of pipe).
b.	Prior to final closure.	Yes (exception: no for closure of support systems or connection of support systems to clean hardware).
c.	During any operations in which a loss of cleanliness could occur that is not easily detected and corrected.	Yes.
d.	Reverification of hardware cleanliness per Section 9.	Yes.
e.	Final inspection following fabrication of hardware or dirt producing modification to hardware.	Yes.
f.	Prior to and during assembly operations (including assembly that creates crevices).	No; however, an independent inspection is required prior to final closures (see inspection b).
g.	Before each thermal treatment.	No.
h.	After each free iron check.	No.
i.	Before and after acid cleaning.	No.
j.	After any test that uses soft metal probes or tips; for example, alloy identity or magnetic particle testing.	No.
k.	After removal of a controlled product as defined in MIL-STD-2041.	Yes, for internal surfaces of hardware that contain unprotected crevices or inaccessible areas.
l.	Following cleaning of internal surfaces.	No; except when performed for a, b, c, d, or e, or when crevices or inaccessible areas of hardware are exposed.
m.	Following tape removal.	No.
n.	Final inspection of cleanliness plug installation.	No, provided the plugs are assembled from clean parts or subassemblies without crevices or inaccessible areas and if the assembly occurs in a clean room or clean area.
o.	During assembly or disassembly of joints with thread sealant downstream of cleanliness filters per 8.8.4.	Yes. Inspector shall monitor joint assembly and disassembly. Where practical, the inspector shall verify that there is no visible thread sealant on internal surfaces following assembly.

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7.2.2 Inspection personnel. Those inspections requiring independent inspection in 7.2.1 shall be performed by personnel not responsible for the cleaning and timely delivery of hardware, unless otherwise approved. Production personnel trained to perform inspections for cleanliness may perform the inspections required by 7.2.1, if an independent inspection is not required.

7.3 Cleanliness inspection methods and acceptance criteria. All hardware shall be cleanliness inspected by one or more of the following methods:

(a) Visual Inspection — Applicable to all internal hardware surfaces that are visually accessible.

(b) Wipe Cloth Inspection — Applicable to physically accessible internal surfaces that cannot be visually inspected.

(c) Swab Inspection — Applicable to piping assemblies without crevices.

(d) Proof of Cleanliness Flush.

7.3.1 Visual inspection. The surface shall be visibly (without magnification) free of foreign material, except:

(a) Small amounts of graphite in isopropanol (where used for assembly) are acceptable.

(b) A few discrete lint fibers per square foot of hardware surface area, each less than 1/4 inch (6.4 millimeters) in length, are acceptable.

(c) Rust in accordance with 7.3.1.1 is acceptable.

(d) Temper films and oxide films in accordance with 7.3.1.3 are acceptable.

(e) Water spots in accordance with 7.3.1.4 are acceptable.

7.3.1.1 Acceptance criteria for rust on new hardware surfaces.

7.3.1.1.1 Corrosion resistant material.

7.3.1.1.1.1 Rust on critical surfaces. A critical surface shall be visibly free of rust. A critical surface exhibiting rust shall be cleaned in accordance with 6.4 to remove the rust and any rust-producing condition. After rust removal, the surface shall be tested in accordance with Test Methods I, II, or III of Appendix B.

7.3.1.1.1.2 Rust on non-critical surfaces. A non-critical surface exhibiting rust shall be cleaned in accordance with 6.4 unless all the following conditions are satisfied:

(a) The rust has no visible thickness.

(b) The rusted area does not exceed one percent of any one square foot (929 square centimeters) area.

7.3.1.1.2 Noncorrosion resistant material.

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7.3.1.1.2.1 Rust. Light superficial rust showing no visible thickness is acceptable on any surface. A tightly adherent oxide deposit that cannot be removed by solvent cleaning is also acceptable. All other rust shall be removed in accordance with 6.4.

7.3.1.2 In-service hardware. For hardware that has been in service and returned for refurbishment, the presence of normally occurring corrosion and wear products are acceptable provided there are no visually apparent unusual debris or substances (determined without magnification). However, readily accessible loose corrosion or wear products should be removed. Except for tanks, filters, and dead legs where sediment may collect, the presence of unusual quantities of wear products and particles exceeding the size criteria of 7.3.4.7, or 7.3.4.8 as applicable, shall be promptly reported to NAVSEA or its authorized representative for information.

7.3.1.3 Metal oxide films. Tightly adherent metal oxide films with no visible thickness that occur on final cleaned surfaces of hardware from specified thermal treatments, are acceptable, except as specified below:

(a) Metal oxide films that can be removed by rubbing with a clean cloth are unacceptable.

(b) Variations in temper film/oxide appearance caused by foreign material are unacceptable. Examples of unacceptable variations include (1) distinct outlines, (2) rough or raised surface areas from tape residue, and (3) dried liquid stains, pools and splashes.

7.3.1.3.1 Abnormal color variations and stains in surface films following thermal treatment.

(a) Abnormal color variations and stains in the oxide film that appear on components following thermal treatment are acceptable provided:

1. The affected surface was cleaned and met the cleanliness visual inspection requirements prior to thermal treatment.

2. There was no loss of cleanliness during thermal treatment or between visual inspection and thermal treatment.

3. The artifact exhibits no visible thickness, no change in surface texture, no raised surfaces, no masking of underlying surface texture appearance, and no evidence of chemical attack based on 1X visual inspection.

4. The artifact cannot be removed by wiping with an acetone wetted cloth.

5. The number of artifacts that could interfere with future cleanliness inspection (that is, large and high contrast artifacts) is limited and there are not abnormal patterns.

(b) All variations in the oxide film that have been accepted per 7.3.1.3.1.a must be documented prior to shipment.

(c) Observed abnormal color variations and stains should be documented for areas that can be visually inspected but cannot be cleaned.

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7.3.1.4 Water spots. Water spots known to result solely from Grade B or better water and are inaccessible for cleaning or where their removal would likely damage the hardware are acceptable.

7.3.2 Wipe cloth inspection. Wipe cloth inspection shall be performed by wiping the surface being inspected with a wet or dry lint-free cloth. Cleanliness shall be evaluated by comparing the wipe cloth with one used to wipe a visually acceptable surface of similar area and finish that has been cleaned in accordance with Section 6. If the wipe cloth exhibits more entrapped or captured foreign material or more discoloration than the comparison cloth, the hardware surface shall be cleaned in accordance with this standard and reinspected. If a cloth is free of discoloration and foreign material, the cloth is acceptable and a comparison cloth is not required.

7.3.3 Swab inspection. Each tube or pipe that is final cleaned by the swab method of A.4 shall be cleanliness inspected in accordance with the following. A comparison swab of the same size as the inspection swab shall be transported through a clean tube of the similar length and diameter. As an alternative, the comparison swab may be rubbed on a visually acceptable surface of similar length. If the inspection swab exhibits more entrapped or captured foreign material or more discoloration than the comparison swab, the tube or pipe shall be cleaned in accordance with this standard and reinspected. If an inspection swab is free of discoloration and foreign material, the inspection swab is acceptable and a comparison swab is not required.

7.3.4 Flushing.

7.3.4.1 Use of flushing. Flushing to restore cleanliness may be performed only when specified in Section 9. Proof of cleanliness flushes may be performed without approval.

7.3.4.2 Flushing procedure. Flushing shall be performed in accordance with a written procedure that includes the following information, as applicable:

(a) Hardware to be flushed.

(b) Type of flush (restoration or proof).

(c) Flushing fluid (grade of water or type of gas).

(d) Flow rate of flushing fluid into the hardware.

(e) Total duration of flush.

(f) Type of effluent filter.

(g) Flow direction or flow path.

(h) Flush equipment.

(i) Flushing filter location(s).

(j) Acceptance criteria (Type I or Type II).

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7.3.4.3 Flushing fluid. Unless otherwise specified, hardware shall be flushed in accordance with the water quality requirements of 8.7. Gas support systems that are disconnected from hardware controlled by this standard, may be flushed with filtered air or filtered inert gas.

7.3.4.4 Flushing flow requirements. Except for flushing of instrumentation in accordance with 7.3.4.9, the following requirements apply to the flow of flushing fluid:

(a) Flow velocity. If practicable, flow velocity during flushing shall equal or exceed the flow velocity during normal operation. If normal flow velocity cannot be achieved during flushing, then, to the maximum practicable extent, flow velocity during flushing shall cause turbulent flow.

(b) Flow direction. Flushing material from other parts of the system into large, low velocity flow regions or into clean portions of the hardware shall be avoided.

(c) Flow volume. The minimum volume of the flush shall be equal to the internal volume of the component being flush tested. For piping systems, at least three times the volume of the system being flushed shall be used. The flow for this volume shall be continuous, if practicable.

7.3.4.5 Filter cloth. A filter cloth's active filtering area shall be no larger than approximately one square foot (929 square centimeters) and shall conform to the technical requirements of MIL-DTL-24671. Detrimental material testing is not required for cloths used for outflushes.

7.3.4.6 Flushing filter inspection and acceptance criteria. After the completion of each flush, the effluent filter shall be visually inspected for cleanliness. Flushing is complete when two separate, successive flushes conform to the acceptance criteria. When flushing used support systems or hardware that has been in service and is returned for refurbishment and the material causing rejection of the hardware is corrosion or wear products, flushing may continue until two successive acceptable flushes are obtained. If unusual debris or substances are obtained on flush cloths, recover per Section 9.

7.3.4.7 Type I flush acceptance criteria. All flushes shall meet Type I criteria unless specified otherwise. The flush is acceptable if the general appearance of the effluent filter cloth is that of a clean white cloth showing no more than slight speckling, soiling, or staining and the material on the filter cloth does not exhibit any of the following:

(a) Fine hair-like sliver larger than 1/16 inch (1.6 millimeters) in any dimension.

(b) Thin flake-like particle larger than 1/16 inch (1.6 millimeters) in any dimension.

(c) Particle of other geometry, larger than 1/32 inch (0.8 millimeters) in any dimension.

(d) Readily apparent (by visual inspection without magnification) quantities of unusual debris or substances, such as resin beads, abrasive grit, oil, grease, brazing flux, welding flux, pipe joint sealer, or preservative.

(e) Visually apparent lint fibers greater than 1/4 inch (6.4 millimeters) in length.

7.3.4.8 Type II flush acceptance criteria. The flush is acceptable if the total quantity of material on the effluent filter cloth is less in area and thickness than a 10 cent piece (0.7 inch

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(18 mm) diameter; 0.05 inch (1.3 mm) thickness) and the material on the filter cloth does not exhibit any of the following:

(a) Fine hair-like sliver larger than 1/8 inch (3.2 millimeters) in any dimension.

(b) Thin flake-like particle larger than 1/8 inch (3.2 millimeters) in any dimension.

(c) Particle of other geometry, larger than 1/16 inch (1.6 millimeters) in any dimension.

(d) Readily apparent (by visual inspection without magnification) quantities of unusual debris or substances, such as resin beads, abrasive grit, oil, grease, brazing flux, welding flux, pipe joint sealer, or preservative.

(e) Visually apparent lint fibers greater than 1/4 inch (6.4 millimeters) in length.

7.3.4.9 Flushing of instrumentation. A disconnected instrument that cannot be flushed by once-through or recirculating methods shall be flushed by either alternate filling and draining or flowing fluid into the instrument through a small diameter tube. Instrumentation shall be flushed until the effluent is visibly clear when observing a static sample or no visible particles or staining are detected when passed through a filter cloth.

8. MAINTAINING CLEANLINESS

8.1 Introduction. This section provides requirements for maintaining cleanliness of final cleaned internal hardware surfaces. Cleanliness is maintained by preventing foreign material from contacting hardware and from entering hardware when it cannot be completely removed.

8.2 Sealing and capping hardware exposed locations. All exposed locations in final cleaned hardware shall be capped, covered, or plugged when not in use. Items used to protect hardware exposed locations, including attached lanyards, shall be examined for deterioration before and after each use. Illustrated examples of temporary caps, seals, and plugs for hardware are shown in Figures 1 through 3. Specific requirements for caps, covers, seals, and plugs are located in Appendix C.

8.3 Cleanliness identity. Shipping documents accompanying hardware that conforms to the requirements of this standard shall state that the hardware meets the cleanliness requirements of MIL-STD-767. If the hardware metal oxide films deemed acceptable per 7.3.1.3 or exhibits water spots deemed acceptable per 7.3.1.4, this shall also be noted in the shipping documents. For hardware fabricated to the requirements of this standard by a sub-tier organization, the organization receiving the hardware shall maintain cleanliness and cleanliness documentation of the hardware in accordance with this standard.

8.4 Tool and equipment cleanliness. Any object or article (such as tools, inspection instruments, hands, and packing and packaging materials) that contacts a final cleaned surface shall be visibly free of foreign material.

8.5 Preventing foreign material from entering hardware. To prevent a loss of cleanliness by the inadvertent introduction of foreign material, the following requirements apply.

8.5.1 Clean room and clean area. The following requirements apply:

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(a) Operations. The following operations shall be performed only within a clean room or a clean area:

1. Assembly of hardware or any portion thereof whenever a final cleaned surface will become an internal surface that is not fully accessible for cleaning or visual inspection.

2. Disassembly of final cleaned hardware with inaccessible areas, unless the hardware will be completely disassembled and re-cleaned.

3. Inspection of internal surfaces of final cleaned hardware with inaccessible areas.

4. Breaking any seal to an inaccessible area of final cleaned hardware.

(b) Cleanliness maintenance of a clean room or clean area. The clean room or clean area and all clean room or clean area equipment, furnishings, and atmosphere shall be maintained clean and free of foreign material and unnecessary parts, fittings, and tools to the extent necessary to ensure that hardware cleanliness is maintained. The clean room and clean area shall be maintained in a condition as free of dirt as is practicable except for that being produced by the particular manufacturing operation in progress. Each clean room or clean area shall be visually inspected and cleaned as required to maintain cleanliness.

(c) Personnel. Personnel working in a clean room/clean area shall wear clean head covers and overalls or smocks, whenever such clothing is necessary to maintain acceptable cleanliness of the hardware. Eating, chewing gum, or use of tobacco in any form (including e-cigarettes) shall not occur within a clean room/clean area. Drinking water shall be confined to restricted areas within the clean room/clean area.

(d) Clean room construction. A clean room shall be a complete, permanent enclosure. Clean room floors, walls, and ceilings shall be hard, smooth, and non-flaking and non-peeling. A clean room door shall be tight fitting unless a positive pressure is created by the clean room ventilation system.

(e) Clean area construction. If appropriate for the hardware and the operation, a clean area may consist of a clean plastic bag or sleeve around appropriate parts of the hardware, a clean tent, a partitioned and covered area, or a clean table in an office. All items used in clean area construction shall be designed to prevent their entry into the hardware. For large hardware, a clean area boundary shall be established to prevent contamination through each opening in use. The clean area boundary need not encompass the entire hardware perimeter and multiple clean area boundaries may be established for a single part. Work on one portion of the hardware may not be performed such that it compromises the cleanliness of another portion of the hardware.

(f) Control of outside foreign material. If needed to control clean room cleanliness, measures shall be taken to prevent foreign material entry via shoes, such as a clean floor covering or clean shoe covers.

8.5.2 Accountability. Accountability provides controls to avoid loss of tools and equipment in hardware with exposed locations. Unless all internal surfaces can be visually observed and certified to be clean prior to closure of the hardware, the following requirements apply:

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(a) A physical boundary shall be established that is located to prevent inadvertent introduction of foreign material, tools, and other equipment into the hardware.

(b) A formal written record shall be maintained for material (for example, tools, equipment, temporary plugs and seals, and tape) taken inside the physical boundary that is small enough to fit inside exposed locations in the hardware and does not have a lanyard per 8.5.2.c. The record shall describe the material, when and where it was introduced, and when it was removed. Except for clear parts on eyeglasses and goggles, accountability is required for all clear materials regardless of size. A separate accountability entry for clear parts securely attached to tools (for example, lenses on flashlights, borescopes, videoprobes, vacuum hose extension) is not required provided that the inspection of the tool verifies that the clear parts remain intact and have not been left in the system.

(c) To the maximum extent practical, each object used within the physical boundary that is small enough to fit into any exposed locations in the hardware shall be securely attached with a clean lanyard to the user, an object larger than the hardware exposed location, or a rigid structure. Lanyards shall conform to the following:

1. Material. Lanyard material shall be corrosion resistant and flexible. Examples include nylon cord, wire rope of a corrosion resistant material, or chain of corrosion resistant material. A lanyard shall not be coated and shall not contain exposed material that chips, flakes, or peels.

2. Strength. A lanyard shall remain intact under the load imposed by the attached object if it were to free fall twice the full length of the lanyard.

3. Length. A lanyard shall be the shortest practicable length consistent with proper use of the attached object.

(d) The accountability record shall be reviewed prior to final closure of the hardware opening. If an object cannot be accounted for, hardware cleanliness has been lost and recovery from a loss of cleanliness shall be conducted in accordance with Section 9.

8.5.3 Shop equipment. Shields and covers (such as drip pans and splash covers) suitable to prevent uncontrolled release of foreign materials shall be affixed as necessary to all shop equipment (such as cranes and hoists) when located above or near an exposed location.

8.5.4 Shop structures. All structures (such as platforms and walkways) above or near an exposed location shall be maintained free of all loose objects and materials or shall be suitably shielded as necessary to prevent a loose object or material from entering hardware.

8.5.5 Hammers. Hammers and mallets used near an exposed location shall not contain loose material such as metal shot.

8.5.6 Special tools. All tools for an application near an exposed location shall be designed and constructed to be "fail-safe"; that is, each part that could fail in use shall be captive so that it will not separate from the rest of the tool if failure does occur.

8.5.7 Other methods of protection. If it is not possible to seal all final cleaned surfaces, crevices, or inaccessible areas in accordance with 8.2 during a specific, unique dirt producing operation, then another method of protection, when approved by the organization’s engineering

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personnel, shall be used to prevent foreign materials from contacting these hardware surfaces. Other methods, for example, may include one or more of the following:

(a) Pressurizing the hardware with clean gas (see 8.8.2.2) at a pressure and flow rate sufficient to prevent entry of foreign material.

(b) Machining without coolant.

(c) Using suction devices in close proximity to the location where the foreign material is generated.

(d) Orienting the hardware so that foreign material will not fall into any unsealed opening.

8.5.8 Welds.

(a) Pipe welding in accordance with NAVSEA 250-1500-1 or NAVSEA S9074-AR-GIB-010/278 of pipe that is verified clean prior to welding is not a dirt producing operation to the internal surfaces of the joined pipe, provided that no events that may compromise cleanliness (such as burnthrough or blowout) occur.

(b) Gas Tungsten Arc (GTA) welding on any surface may be considered a non-dirt producing operation.

(c) For carbon steel and copper-nickel socket welds, groove welds made with backing strips or rings, partial penetration welds, fillet welds, and pipe sleeve-to-pipe welds performed without a purge gas, the backside of the weld may develop a loose oxide film and shall be mechanically cleaned whenever the backside is physically accessible for cleaning without disassembly. Cleaning need not occur if the backside of the weld is not physically accessible for cleaning without disassembly.

8.5.9 Clean pipe cuts. A clean pipe cut is a potentially dirt producing operation but does not require restoration of cleanliness per Section 9 provided the following requirements are met:

(a) The cut is performed with a method qualified on hardware of similar size and material to not introduce foreign material. During qualification, the hardware being cut shall be completely visually inspected to demonstrate no foreign material was introduced.

(b) The cut is conducted by personnel qualified to perform this method.

(c) The hardware is verified clean after the cut by visual inspection to the maximum extent practical without special equipment.

(d) Whenever practical, cut locations should be selected on horizontally oriented pipe to facilitate inspection.

(e) If inspection indicates entry of foreign material into clean hardware, restore cleanliness per Section 9.

8.6 Tape. Use of tape shall comply with all of the following:

(a) Tape shall be visible without mechanical aids (such as borescope or mirror(s)).

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(b) Tape shall be colored and exposed only to temperatures below the tape's temperature rating. Transparent or metal colored tape is prohibited. Yellow and/or magenta colored tape is prohibited for hardware that is being shipped.

(c) Tape alone shall not be used to physically prevent cleanliness plug or seal movement into inaccessible areas of hardware.

(d) Tape alone shall not be used as a cleanliness barrier, except for the following:

1. The opening to be covered is no larger than 1-inch diameter and should not contain any sharp edges.

2. The size of tape is at least 2-inch by 2-inch, extends at least 1/2-inch beyond the opening, and is securely affixed to the surface.

3. The tape is not being used for shipped hardware, must contain at least two layers, and be readily visible throughout the entire duration of its use as a cleanliness barrier (for example, will not be covered by other boundaries or parts before it is removed).

4. The hardware is not moved while tape alone is being used as a cleanliness barrier.

5. The openings that use tape alone and are greater than 1/2 inch in diameter shall be documented on a log and inspected for integrity at least weekly while installed.

(e) Tape may be used to seal the crevice between a cleanliness plug or seal and hardware surface provided all internal crevices and inaccessible areas on the taped side of the plug or seal are sealed to prevent tape entry.

(f) Tape may be used where it is the only practicable means of accomplishing the work, provided all internal crevices and inaccessible areas are sealed from the areas where tape is used.

(g) After tape is removed from hardware surfaces, the surfaces shall be visually inspected and cleaned, as necessary, to ensure all tape reside has been removed.

(h) Any other use of tape must be approved.

8.7 Water requirements.

8.7.1 Water purity. Three grades of water, A, B and C in descending order of purity, may be specified for cleaning, flushing, and testing. Water purity requirements specified throughout this standard are minimum requirements and do not preclude the use of water of greater purity. The requirements for Grades A, B and C water are listed in Table IV.

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TABLE IV. Water purity.

Water attribute	Water Grade		
	A 1/	B	C 2/
Chloride ion, ppm (maximum) 3/	0.1	1.0	35
Conductivity, micromhos/cm (maximum) 4/ 5/	2.5	20	550
Resistivity, minimum, ohms-cm 4/	400K	50K	1.8K
pH (range) 6/	5.7 - 8.0	---	---
Visual clarity	No turbidity, oil, sediment, or resin beads		

1/ Demineralized water meeting the requirements of ASTM D1193, Type II, satisfies Grade A requirements.
2/ Potable water and municipal water generally meet Grade C requirements, but still must be confirmed via appropriate testing.
3/ Chloride ion need not be measured if the conductivity of Grade B or C water is less than 2.5 micromhos/centimeter, or if the resistivity is 400,000 ohms-cm or greater.
4/ Measure either conductivity or resistivity; resistivity is the inverse of conductivity.
5/ Microsiemens/cm and micromhos/cm are equivalent units.
6/ pH need not be measured if the conductivity of Grade A water is 0.25 micromhos/centimeter or less, or if the resistivity is 4,000,000 ohms-cm or greater.

8.7.1.1 Minimum water sampling test frequency. Each water source (that is, public utility, private ground water supply, bottled water vendor, and organization's water purification equipment and support systems) shall be sampled and tested for conformance to Table IV at a frequency sufficient to ensure the water quality specifications are not violated. The frequency shall be specified in the organization's cleanliness control procedures.

8.7.2 Water grade requirements for testing. Table V specifies the minimum grades of water that shall be used for tests of hardware, unless otherwise specified by other contractual documents.

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TABLE V. Required grade of test water.

Test temperature, °F (°C)	Hardware material	Water grade
Less than 150 (66)	All, except nickel-chromium-boron (Colmonoy) alloys	C 1/
Less than 150 (66)	Nickel-chromium-boron (Colmonoy) alloys	B
150 (66) or above	All, except copper alloys and copper-nickel alloys	B
150 (66) or above	Copper alloys and copper-nickel alloys 2/	C 1/

1/ Grade C water shall be removed from the hardware as soon as practicable after completion of the test and the hardware shall be rinsed with Grade B or better water.
2/ All wetted hardware surfaces shall be copper or copper-nickel.

8.7.3 Water grade requirements for contact with a final cleaned surface. Unless otherwise specified, if water is used in contact with a final cleaned surface, Grade B or better water shall be used as a final rinse, and hardware shall be dried following contact.

8.8 Support system requirements.

8.8.1 Support system cleanliness. Unless otherwise approved by NAVSEA, support systems (see 3.23) shall conform to the cleanliness requirements of the hardware to which they will be temporarily connected. Support systems shall also meet the detrimental material control requirements of MIL-STD-2041 to the extent necessary to prevent detrimental materials from being introduced into hardware. For water support systems, the detrimental material control requirements of MIL-STD-2041 shall be applied as necessary upstream of the 80-micron cleanliness filter (if installed) to preclude transfer of dissolved detrimental materials and impurities. In addition, any nonmetallic surfaces upstream of the 80-micron cleanliness filter (if installed) with sufficient surface area to leach impurities into the water (i.e. greater than approximately 5 square inches of wetted surface area for an individual part) shall not contain halogens (except for polytetrafluoroethylene) or sulfur in their chemical structure. These requirements need not apply if the water entering the hardware can be shown to meet Grade B or better via continuous monitoring during hardware testing and automatic shutoff or alarm exists when limits are exceeded. Support system cleanliness and detrimental material controls need not apply if the hardware will not be exposed to temperatures greater than 350°F during the testing and will be subsequently disassembled entirely to piece parts (that is, with no crevices or inaccessible areas remaining), all of which will be cleaned and inspected upon completion of the testing.

8.8.2 Cleanliness boundary. If the entire support system is not maintained clean, then a cleanliness boundary shall be established as follows:

8.8.2.1 Water system. A nominal rating size 80 micron (or finer) metal filter shall be used. Cleanliness shall be maintained between the hardware and the metal filter.

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8.8.2.2 Gaseous system. A nominal rating size 80 micron (or finer) metal filter downstream of a nominal rating size 10 micron (or finer) fiber filter capable of removing oil and foreign material shall be used. Cleanliness shall be maintained between the hardware and the metal filter.

8.8.2.2.1 Weld gas. Support systems that deliver weld shield gas or purge gas that contacts final cleaned internal surfaces of hardware, shall either conform to the support system requirements in 8.8.2.2 or all of the following:

(a) Purchase the gas to an applicable specification with minimum purity requirements.

(b) Use a nominal rating size 80 micron (or finer) metal filter in the support system; and maintain cleanliness between the hardware and the metal filter.

(c) Prior to welding or daily, whichever is less frequent, purge the weld gas through a clean lint-free cloth; and subsequently verify that the cloth is visibly free from oil and other foreign material.

8.8.2.3 Filter cleaning. Temporary support system cleanliness filters that have not been fabricated to this standard shall be cleaned by ultrasonic cleaning (6.3.2) or another approved method and then flushed to a type I acceptance criteria (see 7.3.4.7).

8.8.2.4 Vent and drain systems. Where open vents or drains could result in introduction of foreign material, an 80 micron (or finer) metal filter or screen shall be installed. Where backflow is not possible, a filter or screen is not required as a cleanliness boundary. Minor backflow of air at low flow rates that is not expected to introduce foreign material need not be filtered (such as inflow of air due to ambient temperature changes).

8.8.2.5 Restrictions. Metal filters shall be either edge-type or mesh-type. A fiber, sintered metal, or high efficiency particulate air filter shall not be used as a cleanliness boundary, or in clean portions of the support system unless upstream of an 80-micron or finer metal filter.

8.8.2.6 Filter and screen inspection. Filters and screens should be inspected, cleaned, or replaced periodically to ensure proper operation.

8.8.2.7 Filter and screen strength. Metal filters and screens shall be capable of withstanding at least the maximum system differential pressure obtainable with the filter clogged. Fiber filters must be able to withstand normal operation differential pressure at the particular location of the support system in which they are installed.

8.8.3 Cleanliness verification of support systems. The support system shall be verified clean by visual inspection, wiping, or flushing, as appropriate. When a cleanliness boundary is installed, the requirements for cleaning and cleanliness verification apply to the surfaces of the support system that contact system fluid between the cleanliness boundary and the hardware to which the system is connected. Support systems shall be either maintained clean or verified clean prior to use.

8.8.4 Thread sealants and other joint compounds. Thread sealants and other joint compounds meeting the detrimental material requirements of MIL-STD-2041 may be used in any support system location upstream of an 80 micron or finer metal filter. Polytetrafluoroethylene (Teflon) jam nuts may be used both upstream and downstream from the

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filter provided they do not degrade in the conditions intended for use and the pipe end is cleaned after inserting the jam nut. Thread sealants may be used in support systems downstream of an 80 micron or finer metal filter provided they meet the following conditions:

(a) Thread sealants shall meet the requirements of MIL-STD-2041, except as permitted by 8.8.1. Use of Teflon tape may be requested on a case-basis for NAVSEA approval.

(b) The thread sealant should provide a better leak-tight seal than Teflon jam-nuts.

(c) Locations where thread sealant may be used shall be specified on applicable tooling drawings and/or schematic sketches. Sealed joints shall be identified by tagging and sealing using tamper proof tape or paint.

(d) Affected support system joints using the thread sealant shall be assembled prior to connection to clean hardware.

(e) Thread sealant shall be applied sparingly to the male end of the fitting and shall not be applied to the thread closest to the fluid.

(f) The proper application of the thread sealant shall be specified in an internal procedure that shall include a proficiency demonstration.

(g) A quality assurance representative (or equivalent) shall monitor joint assembly and where practical verify that there is no visible thread sealant product on internal surfaces following assembly.

(h) Joints containing thread sealant shall be distinctively marked and shall not be disassembled while the support system is connected to clean hardware to preclude contaminating the hardware with thread sealant residue.

(i) For any support system to be reused, a quality assurance representative (or equivalent) shall be present for joint disassembly to ensure cleanliness requirements are met.

8.8.5 Support system openings. All openings in each support system being maintained clean shall be sealed in accordance with 8.2 and labeled with a sticker or other device near the sealed opening to identify the condition of cleanliness. If such a sticker or other device is discovered missing, no violation of cleanliness requirements has occurred provided knowledge or evidence (for example, sticker adhesive residue, reliable personal observation, written record) exists that the support system was originally clean and the cleanliness plug or cover appears intact and undisturbed. If these conditions are met, affix a new sticker; otherwise, take action to verify the cleanliness of the support system (8.8.3).

8.8.6 Support system hose inspection. Each hose in support systems (other than fluid discharge) shall be visually inspected periodically and shall be replaced if there is evidence of deterioration such as cracking, chipping, or flaking.

8.8.7 Pump system cleanliness. Pumps (such as hydrostatic test pumps) used in support systems and lubricated by a fluid other than the pumped fluid shall be of the type that employ an air gap or other design feature which prevents contamination of the effluent with lubricating or actuating fluids. This requirement is not applicable to vacuum pumps. However, evacuation

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systems should be designed to prevent foreign material from entering hardware in the event the vacuum pump should fail.

8.8.8 Support system lubricant. In addition to graphite in isopropanol and Grade B water, ACS reagent grade glycerol or silicone compound per SAE AS8660 may be used for lubricating O-rings and other fit-up applications in support systems. The lubricant selected should not degrade the O-ring. Additionally, commercial quick-disconnect fittings with silicone lubricants that do not meet SAE AS8660 may be used provided that the following conditions are met:

(a) Experience indicates that disassembly of the fitting for cleaning and application of an authorized lubricant will likely damage the fitting.

(b) The silicone lubricant meets the shop product detrimental material requirements of MIL-STD-2041.

(c) Only a thin film of lubricant is present on the internal surfaces of a representative disassembled quick-disconnect fitting. The organization's engineering personnel shall determine when inspections are necessary. At a minimum, inspections shall be performed for new fitting designs or new vendors.

(d) Ultrasonic cleaning or flushing of the quick-disconnect fitting is performed prior to initial use.

(e) The silicone lubricant is not soluble in water.

(f) Applicable requirements for inspections and use of quick-disconnect fittings with silicone lubricants in support systems shall be included in an internal procedure.

9. RECOVERY FROM A LOSS OF CLEANLINESS OR LOSS OF CLEANLINESS CONTROL

9.1 Procedure. Recovering from a loss of cleanliness or loss of cleanliness control involves the following steps:

(a) Inspection for foreign material

(b) Removal of foreign material detected

(c) Verification of cleanliness

9.2 Inspection for foreign material.

(a) The organization shall attempt to determine the location of the foreign material. If no foreign material is found and the inspection meets the criteria for verification of cleanliness (see 9.4), cleanliness has not been lost and no additional actions are needed.

(b) If foreign material found is also a detrimental material, the recovery procedure of MIL-STD-2041 shall also be followed.

9.3 Removal of foreign material. After detection of foreign material in hardware, the following removal requirements apply:

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(a) A method known to be effective shall be used to remove the foreign material. For simple cases, this could merely be tool retrieval, wiping, or vacuuming.

(b) Flushing to remove foreign material is allowed only when approved, when authorized by NAVSEA approved manuals or procedures, or as allowed below. When performed, the flushing method and medium shall be known to be effective for removal of the type of foreign material to be removed.

1. Support systems and hardware that do not contain unflushable dead legs or low velocity flow regions may be flushed to reestablish cleanliness. Support systems shall be disconnected from hardware before flushing.

2. For support system components that cannot be purchased clean to this standard (such as commercial off the shelf items), where component disassembly is not easily accomplished, and flushing is expected to be effective, cleanliness may be established by flushing prior to installation in a support system.

3. Gages that have been removed from installed hardware may be flushed to establish cleanliness (see 7.3.4.9).

4. Piping systems without branches and dead legs may be flushed to reestablish cleanliness.

(c) If the removal methods covered by 9.3.a and 9.3.b are not expected to be effective, the hardware shall be disassembled and cleaned, or an alternate recovery method shall be approved. Such removal methods must be known to be effective for removal of the type of foreign material present on the hardware based on mockup, testing, or past experience.

(d) Re-establishment of cleanliness does not require approval provided:

1. The cleaning method is in accordance with Section 6. (Cleaning Requirements),

2. The cleanliness of inaccessible areas and/or crevices is not compromised, and

3. Cleanliness is verified (see 9.4).

9.4 Verification of cleanliness. Following removal of all foreign material detected, cleanliness shall be verified as follows unless all foreign material that entered the hardware has been fully recovered:

9.4.1 Hardware without inaccessible areas and crevices. Hardware shall be inspected using either a visual inspection (7.3.1) or wipe cloth inspection (7.3.2) or for pipe, swab inspection (7.3.3).

9.4.2 Hardware with inaccessible areas and crevices. All accessible surfaces shall be inspected in accordance with 9.4.1. Inaccessible surfaces and crevices may be flushed (7.3.4) to verify cleanliness provided the foreign material suspected to be present in the crevice can be removed by flushing (for example, metal particles from cutting or machining; lapping grit). If an alternate cleaning method (9.3.c) is recommended, the verification of cleanliness method shall be recommended at the same time. Verification of cleanliness of crevices and inaccessible areas is not required when either conditions (a) and (b) or conditions (a) and (c) are met:

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(a) Any entry of foreign material is limited in nature (for example, a small quantity of particles past a deficient cleanliness seal; low energy pipe penetration), no foreign material was introduced at high energy (for example, not a grind through) and no operation (such as purging, venting, excessive vibration, or hardware movement) has taken place that could spread the foreign material.

(b) The geometry and visual inspection criteria of (1) or (2) below are met.

1. The pipe is completely inspected and cleaned up to verifiable blockages (such as shut valves or blanked piping) in all directions from the opening.

2. Piping is completely cleaned and inspected to the maximum extent practical but as a minimum shall meet the following criteria:

i. For horizontal or upwardly sloping pipe, three pipe diameters or three inches (7.6 cm), whichever is greater beyond the opening, or if metal chips or slivers were found, the same distance beyond the furthest metal chip or sliver.

ii. For piping that slopes downward to a horizontal or upwardly sloping section, three pipe diameters or 12 inches (30.5 cm), whichever is greater, beyond the low point (or start of the horizontal run of pipe) or, if metal chips or slivers were found beyond the low point, the same distance beyond the furthest metal chip or sliver.

iii. There are no crevices or areas inaccessible to visual or wipe cloth inspection in the piping sections inspected per the above.

(c) Limited loss of cleanliness or loss of cleanliness control. If the visual inspection requirements of 9.4.2.b cannot be met (such as for a component) for a loss of cleanliness, the vendor's engineering personnel may approve cleanliness verification by an alternate inspection meeting the technical intent of 9.4.2.b provided that (1) all evidence (for example, location and size of foreign material relative to inaccessible areas; method of foreign material entry) indicates foreign material was not introduced into crevices or inaccessible areas and (2) the inspections are sufficient to conclude that no foreign material likely remains. For these cases, the details and justification shall be reported to NAVSEA or its authorized representative for information as limited loss of cleanliness or cleanliness control.

9.4.3 Unauthorized lubricants. In the case of unauthorized lubricants in air system joints, visually inspect at least three pipe diameters from the joint to confirm no migration of lubricant into the system and analyze to confirm that the lubricant does not contain carbon.

9.5 Records. The introduction into final cleaned hardware of any foreign material (other than a known number of discrete objects that can be promptly and completely removed), shall be locally documented.

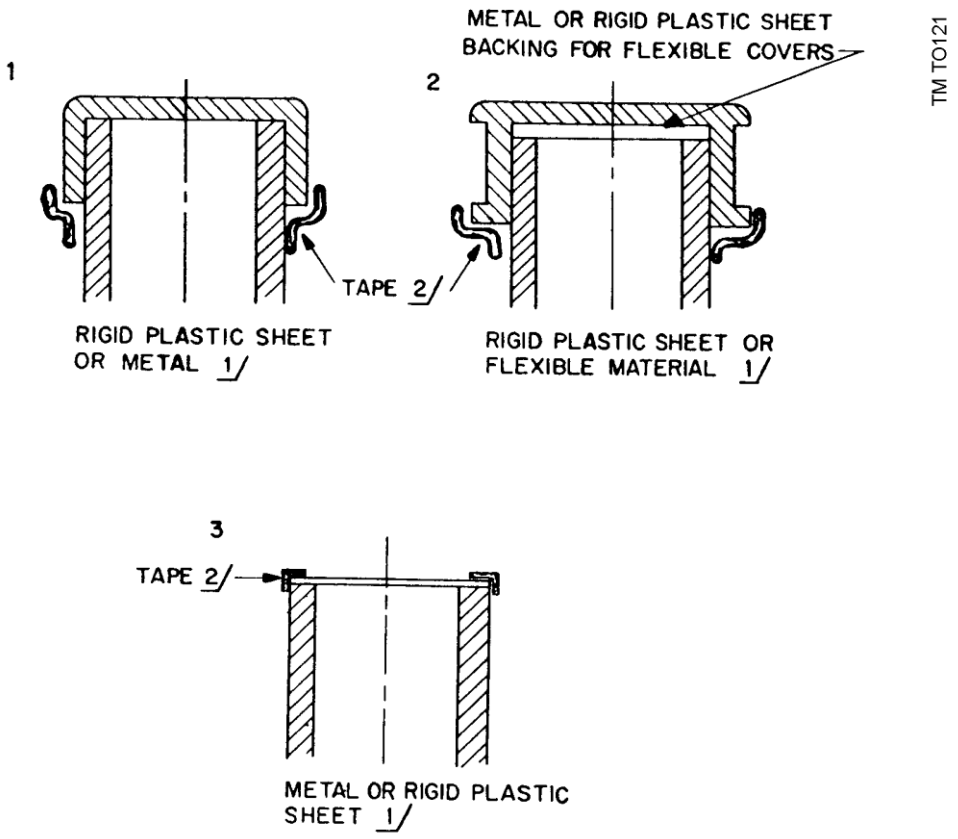
9.6 Hardware acceptance. If cleanliness cannot be restored to hardware in accordance with Section 9, formally submit a recommendation for approval to accept hardware.

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10. NOTES

10.1 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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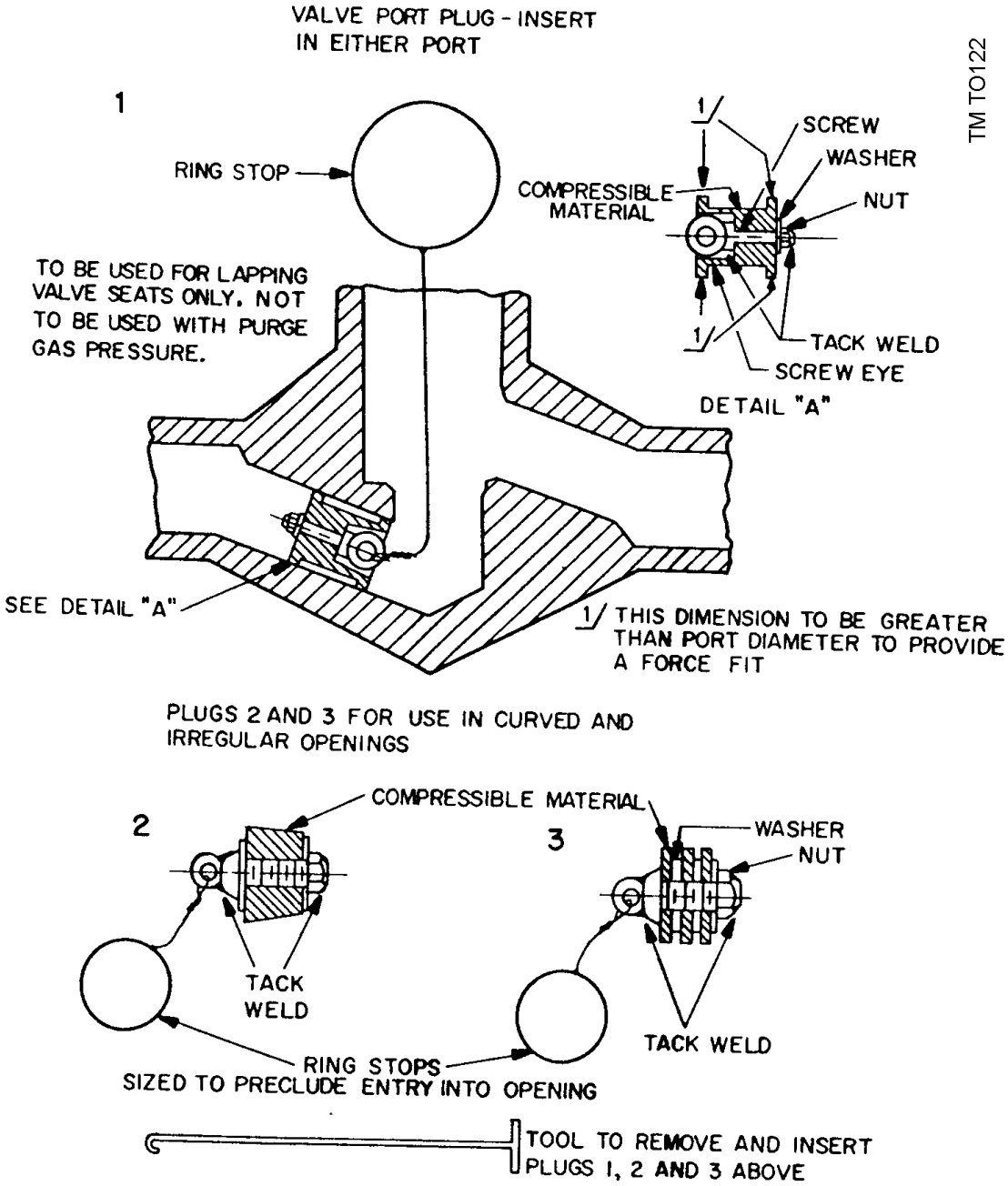
NOTES:

1/ Materials used shall meet the requirements of Appendix C.

2/ Tape in accordance with 8.6 may be used as necessary to ensure that caps will remain on pipe.

FIGURE 1. Nonexpandable caps and seals.

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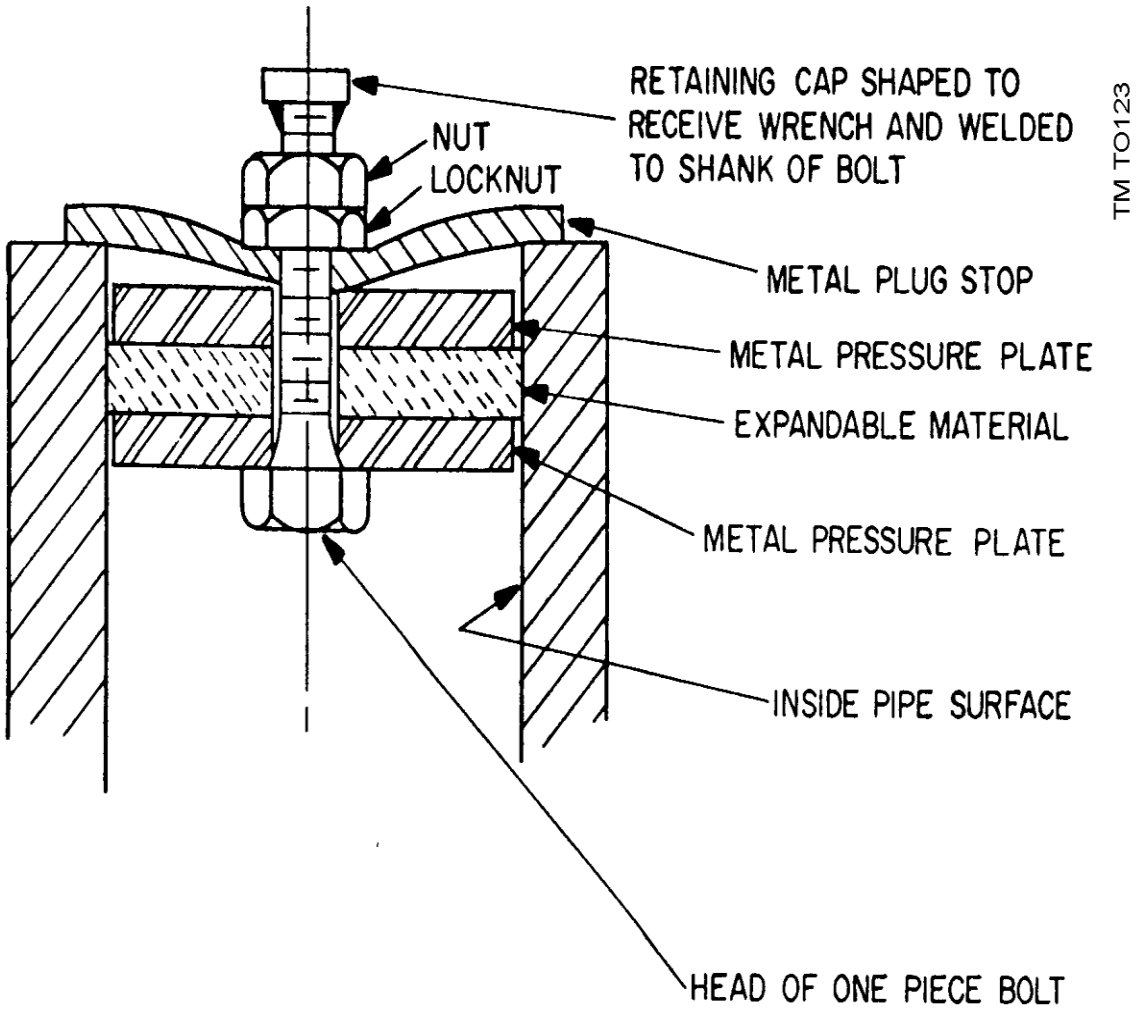


NOTES:

1/ Materials used shall meet the requirements of Appendix C.

FIGURE 2. Nonexpandable plugs.

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NOTES:

1/ Materials used shall meet the requirements of Appendix C.

FIGURE 3. Expandable plug.

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CLEANING METHODS

A.1 SCOPE

A.1.1 Scope. This appendix specifies methods for removal of rust, cleaning tube and pipe, acid cleaning, blast cleaning, tumbling, and removal of tape. This Appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section apply to the extent specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

A.2.2 Government documents.

A.2.2.1 Specifications. The following specifications form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

FEDERAL

A-A-55828	-	Sulfuric Acid, Technical
A-A-59105	-	Nitric Acid, Technical
A-A-59123	-	Sodium Dichromate, Dihydrate, Technical
A-A-59370	-	Ammonium Hydroxide, Technical
A-A-59563	-	Sodium Carbonate, Anhydrous, Technical

MILITARY

MIL-I-17433	-	Inhibitor, Hydrochloric Acid Descaling and Pickling Solutions
MIL-DTL-24641	-	Acid, Hydrofluoric, Technical
MIL-DTL-32594	-	Sodium Phosphate, Tribasic, Technical; Anhydrous, Dodecahydrate, and Monohydrate
MIL-PRF-680	-	Degreasing Solvent

(Copies of these documents are available online at https://quicksearch.dla.mil.)

A.2.3 Non-Government publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL (ASTM)

ASTM E3	-	Standard Guide for Preparation of Metallographic Specimens

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Check the source to verify that this is the current version before use.

ASTM E1146 - Standard Specification for Muriatic Acid (Technical Grade Hydrochloric Acid)

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL (SAE)

SAE J314 - Felts - Wool and Part Wool

(Copies of this document are available online at www.sae.org.)

A.3 RUST REMOVAL PROCEDURE

A.3.1 Procedure. The following procedure may be specified to remove rust from carbon and low alloy steels.

(a) Remove grease by degreasing in accordance with Section 6.

(b) Remove as much rust as is practicable by mechanical cleaning in accordance with Section 6.

(c) If any rust remains after completion of step b, remove remaining rust with rust stripper as follows:

1. Spray, brush, or swab with rust stripper for 2 to 7 minutes. (NOTE: Rust stripper shall consist of concentrated phosphoric acid [at least 68 grams per 100 milliliters of Grade B or better water], diluted with 3 volumes of room temperature Grade B or better water. Optionally, butyl cellosolve [40 milliliters] and non-ionic detergent [5 ml] may be added.)

2. Rinse with Grade C or better water before the surface dries.

3. Rinse with neutralizing solution as specified by A.5.7.1 before the surface dries (optional).

4. Rinse with Grade B or better water before the surface dries.

5. Dry in accordance with 6.6.

A.4 TUBE AND PIPE CLEANING PROCEDURE

A.4.1 Procedure. The following procedure may be applied to final clean surfaces of tube or pipe that contain no crevices or inaccessible areas.

(a) Insert a clean, unused swab conforming to A.4.2 into one end of the tube/pipe.

(b) Move the swab through the length of the tube/pipe by pulling with a lanyard in accordance with A.4.3 or by blowing with pressurized gas in accordance with A.4.4. If the entire length of the tube/pipe cannot be verified free of entrapped swabs, then a lanyard shall be used.

(c) Evaluate the swab in accordance with 7.3.3.

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(d) When the swab is acceptable per 7.3.3, then dry the tube/pipe using a clean, dry swab.

A.4.2 Swab requirements. The swab shall conform to the following:

(a) The swab shall consist of a wool felt pellet conforming to SAE J314 Type F-1 or F-10 or equivalent, or a piece of clean, lint-free cloth.

(b) The swab shall be sized to contact the tube/pipe internal diameter.

(c) The swab shall be saturated with Solvent A, B, C, or D, (see Table I) as applicable, in accordance with the requirements of Section 6.

A.4.3 Lanyard requirements. The lanyard material and cleanliness shall conform to the applicable requirements of 8.5.2.c.

A.4.4 Pressurized gas requirement. Pressurized gas systems shall meet the applicable support system requirements in 8.8.

A.5 ACID CLEANING PROCEDURE

A.5.1 General. Acid cleaning shall be performed in accordance with the following requirements.

A.5.2 Order of precedence. When these requirements for acid cleaning conflict with the requirements of another applicable contract document, the acid cleaning requirements of the other applicable contract document shall take precedence.

A.5.2.1 Use of acid cleaning. Acid cleaning shall be used only after other cleaning methods have been determined to be impracticable or incapable of removing the detrimental or foreign material involved. The following items shall not be acid cleaned:

(a) Porous materials (such as chromium plating, sintered metal carbide, carbon-graphite materials, and alumina).

(b) Nitrided surfaces.

(c) Bearings and other wear surfaces.

(d) An alloy not listed in Table A-I.

(e) Hardware with crevices or inaccessible areas.

(f) Final surfaces of seal membranes.

A.5.3 Acid cleaning procedure submittal and content. Each acid cleaning procedure shall be submitted for approval. The acid cleaning procedure shall contain the following information:

(a) Identification of hardware and material(s) to be acid cleaned.

(b) The thermal treatment history of each material involved.

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(c) The surface condition of each surface that will contact the acid.

(d) Each foreign or detrimental material to be removed.

(e) The results of all mandatory and optional immersion time evaluations.

(f) Identification of the acid solution and concentration.

A.5.3.1 Procedure for alternate acid cleaning process. If the acid cleaning process will not conform to the applicable parameters for a method specified in Table A-I the acid cleaning procedure shall contain the following information in addition to that specified in A.5.3:

(a) Justification for using alternate acid cleaning process.

(b) The quantity, specification, and concentration of every ingredient of the acid cleaning solution, including water Grade.

(c) The method(s) to be used for monitoring and maintaining the acid cleaning solution, including the concentration range of every ingredient and maximum metal content.

(d) Temperature limits.

(e) Maximum cumulative immersion time.

A.5.4 Solutions. Acid cleaning, rinsing, and neutralizing solutions shall be agitated during use to provide an even flow of solution and to maintain uniform temperature and concentration. The temperature of the acid cleaning solution shall not vary more than 10°F (5.6°C) within the tank. Each acid solution shall be analyzed for the concentration of each applicable ingredient listed in Table A-I, no less often than specified below:

(a) Before the initial use of a new solution.

(b) Continuous use - once every eight hours.

(c) Intermittent use - once every seven days.

An acid solution that does not conform to the parameters specified in Table A-I shall be replenished to conform or be discarded. ACS reagent grade chemicals, if available, may be used in lieu of the specifications listed in Table A-I and A.5.7.1.

A.5.5 Immersion time evaluation. Immersion time is the time interval from the start of immersion in the acid solution to complete immersion in the subsequent rinse. If the hardware to be acid cleaned contains a brazed joint or a weld that the acid will contact, immersion time shall be determined by acid cleaning representative test samples. Immersion time evaluation of non-brazed/welded hardware is optional.

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TABLE A-I. Acid cleaning methods.

Alloy Group	Alloys	Treatment Code	Solution Ingredient	Ingredient Specification/ Requirements	Ingredient Amount 4/	Temperature °F (°C)	Total Cumulative Acid Immersion Time, Minutes
I	AISI Types 304L (UNS S30403) 1/ 304 (UNS S30400) 1/ 308 (UNS W30810) 1/ 309 (UNS W30910) 1/ 316L (UNS S31603) 1/ 316 (UNS S31600) 1/ 317L (UNS S31703) 1/ 347 (UNS S34700) 1/ 348 (UNS S34800) 1/ Precipitation hardening corrosion resistant steel Cobalt-base alloys	A	Nitric acid Hydrofluoric acid	A-A-59105, 1.42 sp. gr. MIL-DTL-24641, 60 percent	15 ± 1 volume percent 1.5 ± 0.1 volume percent	125 ± 5 (52 ± 3)	15 max.
		B	Hydrochloric acid	20° Baume Grade of ASTM E1146	50 ± 2 volume percent	125 ± 5 (52 ± 3)	15 max.
II	AISI 403 (UNS 40300) AISI 410 (UNS 41000)	C	Nitric acid Hydrofluoric acid	A-A-59105, 1.42 sp. gr. MIL-DTL-24641, 60 percent	15 ± 1 volume percent 1.5 ± 0.1 volume percent	125 ± 5 (52 ± 3)	5 max.
III	Nickel-chromium-iron alloys 2/	D	Nitric acid Hydrofluoric acid	A-A-59105, 1.42 sp. gr. MIL-DTL-24641, 60 percent	15 ± 1 volume percent 1.5 ± 0.1 volume percent	85 ± 15 (29 ± 3)	15 max.

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TABLE A-I. Acid cleaning methods – Continued.

Alloy Group	Alloys	Treatment Code	Solution Ingredient	Ingredient Specification/ Requirements	Ingredient Amount 4/	Temperature °F (°C)	Total Cumulative Acid Immersion Time, Minutes
IV	Nickel-base alloys other than those of Group III	E	Sulfuric acid Sodium nitrate	A-A-55828, 1.83 sp. gr. Commercial or industrial grade	8.6 ± 0.5 volume percent 0.5 pound/gallon	185 ± 5 (85 ± 3)	30 max.
V	Copper-base alloys Nickel-copper alloys 3/	F	Sulfuric acid Sodium dichromate	A-A-55828, 1.83 sp. gr. A-A-59123	3 ± 1 weight percent 2 ± 0.2 weight percent	112 ± 12 (44 ± 7)	30 max.
V	Pure copper	H	Phosphoric acid	ACS Reagent Grade	16.5 ± 1 volume percent	70 ± 5 (21 ± 3)	20 max.
VI	Carbon and low alloy steel	G	Hydrochloric acid Inhibitor	20° Baume Grade of ASTM E1146 MIL-I-17433	11 ± 1 weight percent 0.4 ± 0.1 weight percent	135 ± 15 (57 ± 8)	15 ± 5

1/ Materials exposed to temperatures of 800 – 1400°F shall receive Treatment “B.” However, the “A” treatment may be used in these cases if the parts are given a solution anneal by heating to 1950°F and water quenching prior to acid cleaning.
2/ Acid cleaning of Alloy X-750 (UNS N07750) and direct aged Alloy 625 (UNS N06625) final cleaned surfaces is prohibited.
3/ This treatment is for removing thin oxide films where a bright clean surface is desired. Otherwise, treatment in accordance with Group IV requirements shall be used.
4/ For each treatment code, the balance of the solution shall be Grade C or better water.

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A.5.6 Intergranular attack (IGA) test.

A.5.6.1 IGA test specimens. Unless otherwise specified, one IGA test specimen shall accompany each acid cleaning batch of Table A-I, alloy Group I, III, or IV (except nickel-copper alloys) hardware through all acid cleaning process steps. The IGA test specimen shall be of the same alloy, product form, thickness, heat treated condition, and brazed or welded condition as the hardware.

A.5.6.2 Metallographic preparation of IGA specimens. From each IGA test specimen, two cross sections perpendicular to the acid cleaned surface, as specified below, shall be prepared in accordance with ASTM E3:

(a) Brazed or welded specimen — one section shall include the braze joint or weld and the heat affected zone; the other section shall include the edge of the test specimen.

(b) Other specimen — each section shall include an edge of the test specimen.

A.5.6.3 Acceptance criteria for IGA. If the depth of IGA exceeds the applicable maximum limit specified in Table A-II, the batch represented by the test specimen shall be rejected. However, the batch is not rejectable if, without violating a drawing minimum dimension, an amount of surface material is removed from the hardware by mechanical methods so that the IGA depth does not exceed the applicable Table A-II limit.

TABLE A-II. IGA depth limits.

Hardware thickness, inches (cm)	IGA maximum depth limit, inches (cm)
0.025 (0.064) or less	0.001 (0.0025)
Over 0.025 to 0.050 (0.127) inclusive	0.0015 (0.0038)
Over 0.050 to 0.125 (0.318) inclusive	3 percent of thickness
Over 0.125	0.004 (0.010)

A.5.7 Acid cleaning procedure. The following procedure shall be used for acid cleaning.

(a) Immediately before immersion in the first batch associated with the acid cleaning process, ensure that hardware surface is:

1. Visually free of foreign and detrimental material that can be removed by wiping (NOTE: This also applies to the surface of all masking material used).

2. Water break free. (A water break free surface is a flat surface that maintains a continuous water film for at least 30 seconds after rinsing in Grade C or better water cooler than 100°F (38°C). Shorter inspection times may be established for a rounded or irregular surface.)

(b) Immerse hardware completely in the active solution in accordance with the conditions of Table A-I. Tubing with an external diameter less than 1/2 inch shall have most of the acid within the tube replaced at least once every minute during the entire immersion period.

(c) Rinse hardware free of the active solution by immersing the hardware completely in the following solutions as soon as practicable after being removed from the active solution.

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1. Rinse in Grade C or better water.

2. Rinse in neutralizing solution as specified in A.5.7.1. (Rinsing of Table A-I, group VI alloys in neutralizing solution is optional.)

3. Rinse in Grade B or better water.

(d) Dry hardware in accordance with 6.6.

(e) Inspect hardware in accordance with A.5.8.

A.5.7.1 Neutralizing solution. The composition of neutralizing solution shall conform to either a, b, or c below

(a) 5 ± 0.5 volume percent ammonia (A-A-59370 Type I) in Grade B or better water. This solution shall not be used on copper or its alloys (such as brass or bronze).

(b) 0.75 ± 0.1 weight percent trisodium phosphate (MIL-DTL-32594) in Grade B or better water.

(c) 3.75 ± 0.75 weight percent sodium carbonate (A-A-59563 Types I, II, or III) in Grade B or better water.

A.5.8 Acceptance criteria for acid cleaned surfaces. Each acid cleaned surface shall be uniform in appearance and shall be clean. If the acid cleaned surface will be the final surface, it shall conform to the surface finish requirements specified by the applicable contract documents. An acid cleaned surface that exhibits rust shall be dispositioned in accordance with 7.3.1.1.

A.6 BLASTING

A.6.1 General. When approved, shot, sand, grit, or vapor blasting shall be performed in accordance with the following:

A.6.2 Blasting media. Unless otherwise approved, No. 140 or finer alumina grit (less than 0.25 weight percent iron) per American Foundrymen's Society (AFS) shall be used.

A.6.3 Blasting cleanliness. Only one generic alloy group of hardware shall be blasted in the same operation. Blasting equipment shall be cleaned and the blasting media cleaned or changed prior to blasting a different alloy group. The blasting procedure submitted for approval shall include final cleaning methods for removal of particulate matter and acceptance criteria for the hardware. For corrosion resistant materials, after the blasting operation, both the surfaces blasted and the adjacent surfaces shall be cleaned with a clean brush and rinsed with Grade C or better water followed by a rinse with Grade B or better water.

A.7 TUMBLING

A.7.1 General. When specifically approved, tumbling shall conform to the blasting operation cleanliness requirements of A.6.

A.8 TAPE RESIDUE REMOVAL PROCEDURE

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A.8.1 General. The following procedure may be used to remove tape residue from metal and carbon-graphite surfaces. If using only for metal surfaces that are free of crevices and inaccessible areas, the solvent need not be tested in accordance with MIL-STD-2041.

(a) Apply n-heptane or degreasing solvent (MIL-PRF-680 Type I) directly to the tape residue and work by hand with a stainless steel brush or cloth until all of the tape residue is dissolved. Hardware may be immersed in the solvent, if necessary.

(b) Wipe any excess solvent from the hardware surfaces with a cloth. A residual film may still be present on the hardware.

(c) Final clean the hardware by rubbing with a cloth soaked with Solvent A, B, or C. The hardware may be immersed in the solvent.

(d) Dry the hardware in accordance with 6.6.

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FREE IRON DETECTION

B.1 SCOPE

B.1.1 Scope. This appendix specifies four test methods to detect free iron. This Appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

B.1.2 Non-Government publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL (ASTM)

ASTM B733 - Standard Specification for Autocatalytic (Electroless) Nickel-Phosphorus Coatings on Metal

(Copies of this document are available online at www.astm.org.)

SAE INTERNATIONAL (SAE)

AMS2460 - Plating, Chromium

(Copies of this document are available online at www.sae.org.)

B.2 PRE-TEST SURFACE CONDITION

B.2.1 General. The surface to be tested shall be free of all materials, such as petroleum products or tape residue, that could interfere with test results.

B.3 TEST METHOD I: AERATED WATER

B.3.1 Solution. The aerated water test solution shall be 155 ± 5°F (68 ± 3°C) Grade B or better water saturated with air.

B.3.2 Test procedure. The aerated water free iron test shall be conducted as follows:

(a) Apply aerated water by a suitable method to all external critical surfaces and all accessible internal critical surfaces to be tested.

(b) Maintain each surface completely wetted by the aerated water solution for a minimum of one hour.

(c) After the one hour wetting period, wipe each wetted surface with a clean dry cloth, clean absorbent paper, or clean cotton swabs.

(d) Visually inspect each critical surface exposed to the aerated water solution in accordance with B.3.3.

B.3.3 Acceptance criteria. If the tested surface exhibits rust or pitting, clean the surface in accordance with 6.4 to remove free iron and retest in accordance with B.3.

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B.4 TEST METHOD II: WET CLOTH

B.4.1 Test procedure. The wet cloth free iron test shall be conducted as follows:

(a) Saturate a clean, lint-free cloth with Grade B or better water at ambient temperature.

(b) Place the saturated cloth in contact with all external critical surfaces and accessible internal critical surfaces.

(c) Maintain each wet cloth contacting a critical surface wet for a minimum of six hours.

(d) After the six hour wet contact period, wipe each wetted surface with a clean dry cloth, clean absorbent paper, or clean cotton swabs.

(e) Visually inspect each critical surface exposed to a wet cloth in accordance with B.4.2.

B.4.2 Acceptance criteria. If the tested surface exhibits rust or pitting, clean the surface in accordance with the 6.4 to remove free iron and retest in accordance with B.4.

B.5 TEST METHOD III: FERROXYL

B.5.1 Test procedure. The ferroxyl free iron test shall be conducted by performing either the ASTM B733 or SAE AMS2460 ferroxyl method on the hardware surface. Ferroxyl solution shall not be applied to any surface containing a crevice or inaccessible area.

B.5.2 Acceptance criteria. If the tested surface exhibits any blue coloration, clean the surface in accordance with 6.4 to remove free iron and test reagents and retest in accordance with B.5.

B.6. TEST METHOD IV: COPPER SULFATE

B.6.1 Solution. Copper sulfate test solution shall be homogeneous and shall consist of the ingredients listed below in the proportions stated:

a. Grade B or better water, 490 ± 10 milliliters.

b. Reagent grade copper sulfate ($CuSO_4$ $5H_2O$), 7 ± 1 grams.

c. 66° Baume (~93%) reagent grade sulfuric acid (H_2SO_4), 3 ± 1 milliliters.

Copper sulfate solution shall not be applied to any surface containing a crevice or inaccessible area or any surface of carbon or low alloy steel. Copper sulfate solutions shall not be used on final surfaces when MIL-STD-2041 applies unless approved.

B.6.2 Temperature. The test solution and the surface to be tested shall be at ambient temperature.

B.6.3 Solution sensitivity check. Before use on the surface to be tested, the copper sulfate solution shall be checked for sensitivity by applying approximately one milliliter to a

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clean, scale free, rust free, carbon steel surface. If the copper sulfate solution does not deposit copper on the carbon steel surface within six minutes, that batch of solution shall not be used.

B.6.4 Test procedure. The copper sulfate free iron test shall be conducted as follows:

(a). Except as specified in B.6.1, apply copper sulfate solution by a suitable method to the hardware external critical surfaces and accessible internal critical surfaces to be tested. Suitable application methods include swabbing, spraying, brushing, and immersion. Only a limited quantity of the copper sulfate solution sufficient to perform the test shall be taken near exposed locations or internal surfaces of hardware. The procedure for application of the test solution shall provide protection of crevices, inaccessible areas, and other surfaces of hardware from drips, runs and other unintended contact with the solution.

(b) Maintain each surface completely wetted by the copper sulfate solution for a minimum of six minutes.

(c) After the six minute wetting period, wipe each wetted surface with a clean dry cloth, clean absorbent paper, or clean cloth swabs.

(d) Visually inspect each critical surface exposed to the copper sulfate solution in accordance with B.6.5.

B.6.5 Acceptance criteria. If the tested surface exhibits deposited copper, clean the surface in accordance with 6.4 to remove free iron and deposited copper, and retest in accordance with B.6.

B.6.6 Post-test treatment. When the tested surface does not exhibit deposited copper, treat in accordance with the following:

(a) Rinse with grade B or better water.

(b) Rinse with neutralizing solution conforming to Appendix A, A.5.7.1.

(c) Rinse with grade B water.

B.6.7 Post test surface inspection. After completion of this free iron test and post test cleaning, all tested surfaces of hardware shall be visually inspected to ensure they are clean, dry, and free of deposited copper, pitting, etching, cracks or other signs of chemical attack.

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REQUIREMENTS FOR CAPS, COVERS, SEALS, AND PLUGS

C.1 SCOPE

C.1.1 Scope. This appendix provides requirements for caps, covers, seals, and plugs. This Appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

C.2. MATERIAL REQUIREMENTS

C.2.1 Caps, covers, seals, and plugs. Caps, covers, seals, plugs, and similar devices used temporarily to maintain cleanliness of hardware shall:

(a) Comply with the detrimental material control requirements of MIL-STD-2041. The restrictions on use of polyvinylchloride (PVC) sheet or film listed in MIL-STD-2041 apply. PVC is a high chloride material and, therefore, should be used as a cleanliness boundary for hardware surfaces, openings, crevices, or inaccessible areas only in cases where other methods are not practical. Care shall be exercised to ensure that PVC does not enter exposed hardware locations.

(b) Be fire resistant to the extent practical.

(c) Be non-brittle, non-splintering, non-melting in the range of temperatures expected in use.

(d) Be non-tearable (for example, no paper, plastic film, or foam). Polyethylene meeting the requirements of MIL-DTL-24466 is considered acceptable..

(e) Not transfer to or mark hardware (for elastomers and plastics, no exudate when tested in accordance with MIL-STD-2041).

(f) Have any metal parts be corrosion resistant to the extent practical. Caution shall be used when using aluminum and aluminum alloy pipe caps, as corrosion can occur under higher humidity conditions and sharp edges can produce particles which will interfere with subsequent welding.

(g) Show no evidence of deterioration during extended exposure at the most severe service environment expected, including incidental contact from solvents used for cleaning the hardware.

(h) Not result in galvanic or other corrosion at the point of contact.

C.2.2 Plastic sheeting. Plastic sheeting may be used to maintain cleanliness of internal hardware parts prior to assembly or shipment, and of removed internal hardware parts. All plastic sheeting shall be colored. Polyethylene meeting the requirements of MIL-DTL-24466 is considered acceptable.

C.2.3 Cloth. Cloth shall not be used except as bags for encasing inflatable cleanliness plugs.

C.3 GENERAL FABRICATION AND USE REQUIREMENTS

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(a) Plugs shall be easily installed without damage to hardware or the flexible material in the plugs or seals. For example, do not use expandable or press-fit elastomeric plugs on threaded surfaces since the threads may tear the elastomer.

(b) Unless approved, plugs and seals shall have an integral or attached part that prevents it from being drawn into inaccessible areas of the hardware. If this part must be removed to accomplish a task during the work, then the following restrictions apply:

1. The hardware shall be vented to prevent forming a vacuum which could pull the plug inside the hardware.

2. The part shall be fastened to the hardware near the plug and labeled.

3. Immediately after completing the task that required the removal of the part, the part shall be replaced and the tag or label removed.

The above allowance shall only be used when necessary to accomplish a task that cannot be performed with the part attached.

(c) Lanyards, attached to cleanliness plugs, shall be secured to external surfaces with minimal slack when work is suspended and the hardware is unattended. Securing lanyards with minimal slack will aid in the detection of loose or dislodged cleanliness plugs.

(d) Temporary plugs and seals should be designed so that pieces or broken parts will not enter hardware.

(e) Plugs and seals shall be visually free of foreign material prior to insertion into hardware in which the plug or seal is to be used.

(f) Cleanliness plugs and seals shall not be used to contain pressure above the pressure required for gas purging, such as during welding.

(g) Short plugs and seals that are subject to cocking, such as those designed to be inserted through pipe elbows, tees, or valves, should be frequently inspected in place to assure proper alignment and continued sealing.

C.4 SPECIFIC FABRICATION AND USE REQUIREMENTS

C.4.1 Expandable plugs and seals. The following requirements apply to expandable plugs and seals:

(a) Unless otherwise approved, inflatable plugs shall consist of an inflatable inner bladder encased in a strong scuff resistant bag, such as canvas, that will retain the bladder if it bursts. A lanyard from the bag shall be attached to a fixed structure outside the hardware or to a ring of diameter greater than the hardware inside diameter. When two inflatable plugs are used in series for the same seal, the outermost plug may be uncovered if necessary to assure seal tightness provided the area outside the inner plug can be easily inspected and cleaned if the outer plug bursts.

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(b) Inflatable plugs and seals shall be tested for air leakage by immersion in Grade B or better water, or by inflating and soap testing, prior to each use. If soap tested, soap shall be removed prior to use of the plug or seal.

C.4.2 Nonexpandable plugs and seals. The following requirements apply to nonexpandable plugs and seals:

(a) Where the design employs O-rings, the design shall minimize the possibility of the O-rings rolling out of their normal position during plug insertion or removal.

(b) Plastic plugs shall be sufficiently rigid to prevent being pushed or drawn into hardware.

(c) Nonexpandable plugs (such as wafer type plugs) may not always protect hardware from foreign material entry during dirt producing operations. The effectiveness of nonexpandable plugs should be demonstrated before use.

C.4.3 Plug movement. Caution should be taken to prevent movement of plugs after installation in a manner that could permit access for foreign material to enter clean hardware, crevices, or inaccessible areas. If plugs are moved after work area cleanliness is dropped, they shall be removed and accessible areas of hardware inspected for foreign material that could have passed the plug.

C.5 CLEANING OF CLEANLINESS PLUGS

C.5.1 General. Flushing is prohibited for removal of work debris from installed cleanliness plugs. Vacuum cleaning is an acceptable method.

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QUALIFICATION OF ALTERNATE FINAL CLEANING SOLVENTS

D.1 SCOPE

D.1.1 Scope. This appendix specifies the requirements, test methods, and reporting requirements to be satisfied to qualify alternate final cleaning solvents for use on all hardware surfaces or hardware with no crevices or inaccessible areas. This Appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

D.2 ACCEPTANCE CRITERIA FOR QUALIFICATION OF ALTERNATE FINAL CLEANING SOLVENTS

D.2.1 Detrimental material content.

D.2.1.1 Requirements. The candidate solvent shall not contain detrimental elements in excess of the limits prescribed for shop products in MIL-STD-2041.

D.2.2 Volatility.

D.2.2.1 Requirements. For use on all hardware surfaces, the solvent shall have a boiling point of 180°F (82°C) or less. For use on hardware with no crevices or inaccessible areas only, the solvent shall be shown to be effectively removed by wiping with a cloth or by still air drying at ambient temperature for thirty minutes or less.

D.2.3 Residue.

D.2.3.1 Requirement. The solvent shall not contain more than 0.005g of non-volatile material per 100 ml of solvent.

D.2.3.2 Test method. A sample of solvent shall be evaporated and the remaining residue weighed.

D.2.4 Cleaning capability.

D.2.4.1 Requirements. The solvent shall be separately tested for its ability to clean oil-base soils and water-base soils by wiping or immersion. The candidate solvent shall dissolve or otherwise remove visual evidence and any sense of touch of the surface soil.

D.2.4.2 Test method. A small quantity of oil-based lubricant or cutting fluid shall be applied to a clean, metal part with a crevice (as defined in 3.6). The crevice used in this test shall have an opening to depth ratio of 0.2 or less and a depth of at least 0.120 inch (0.30 cm). See Figure D-1 for an example of a crevice test fixture. This part should be prepared such that the crevice can be examined following cleaning and drying (for example, two metal pieces bolted together with the gap between being the crevice). After at least one hour dwell time, the candidate solvent shall be applied to the surface by immersion or saturated wiping cloth and all visible soil removed. Removal of the oil/lubricant shall be confirmed by visual inspection of the external surfaces of the part and examination of the crevice and by touching the surfaces. This procedure shall be repeated with a water-base lubricant or cutting fluid.

D.2.5 Corrosiveness.

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D.2.5.1 Requirements. When applied to the surface of nickel-base, corrosion resistant steel, copper-base and iron-base material, the candidate solvent shall not pit or discolor the surface.

D.2.5.2 Test method. A small amount of the candidate solvent shall be placed on a clean, machined, or ground (surface finish of 250 Ra or better) surface of samples of nickel-base alloy (such as Alloy 600 [UNS N06600]), corrosion resistant steel (such as 304SS [UNS S30400] or 17-4PH [UNS S17400]), copper-base alloy (such as bronze or Alloy 400 [UNS N04400]) and carbon steel. The samples shall be kept warm (80 – 100°F [27 – 38°C]) for two days. Normally the solvent will dry very early in this period. The samples shall then be examined at 20X for any evidence of pitting or surface discoloration.

D.2.6 Environmental, health, and safety considerations.

D.2.6.1 Requirements. No specific requirements are specified. However, when selecting an alternate solvent, all applicable local, state, and federal environmental, health and safety regulations should be complied with. The Safety Data Sheet and any technical literature or reports available from the solvent manufacturer shall be submitted with the reports required by D.4.

D.2.7 Chemistry stability.

D.2.7.1 Requirement. A sample of the candidate solvent should not have a detrimental effect on system chemistry.

D.2.7.2 Test method. Add enough of the candidate solvent to Grade A water to form a one percent solution. Measure and record pH and conductivity. Obtain baseline total organic carbon (TOC) and ion chromatography (IC) analysis (anion, cation and organic acid). Rinse a clean Parr reaction vessel, or equivalent, with Grade A water. Add the one percent solution to the reaction vessel and heat at 500°F ± 20°F (260 ± 11°C) for at least 20 hours. Allow solution to cool and repeat pH, conductivity, TOC and IC analyses on the post heat solution. Record results along with any visual observations regarding the appearance of the solution or condition of the reaction vessel before and after heating the solution.

D.2.7.3 Alternate procedure. As an alternate to D.2.7.2, approximately 16 oz (453 g) may be forwarded to Bettis Laboratory, P.O. Box 79, West Mifflin, PA 15122-0079, Attention: Manager, Water Chemistry, for this test.

D.3 REPORT

D.3.1 Results. The results of the tests covered by D.2.1 through D.2.7 shall be submitted for approval, with a copy to Bechtel Plant Machinery, Inc. (BPMI), 3500 Technology Drive, Monroeville, PA 15146-2459, Attention: Manager, Materials Engineering. The report should also state whether the solvent is to be used for all hardware surfaces or hardware with no crevices or inaccessible areas, and whether it is to be used for oil-base or water-base soils or both. For continued use, specific solvents must be requalified for detrimental material content (MIL-STD-2041) every four years.

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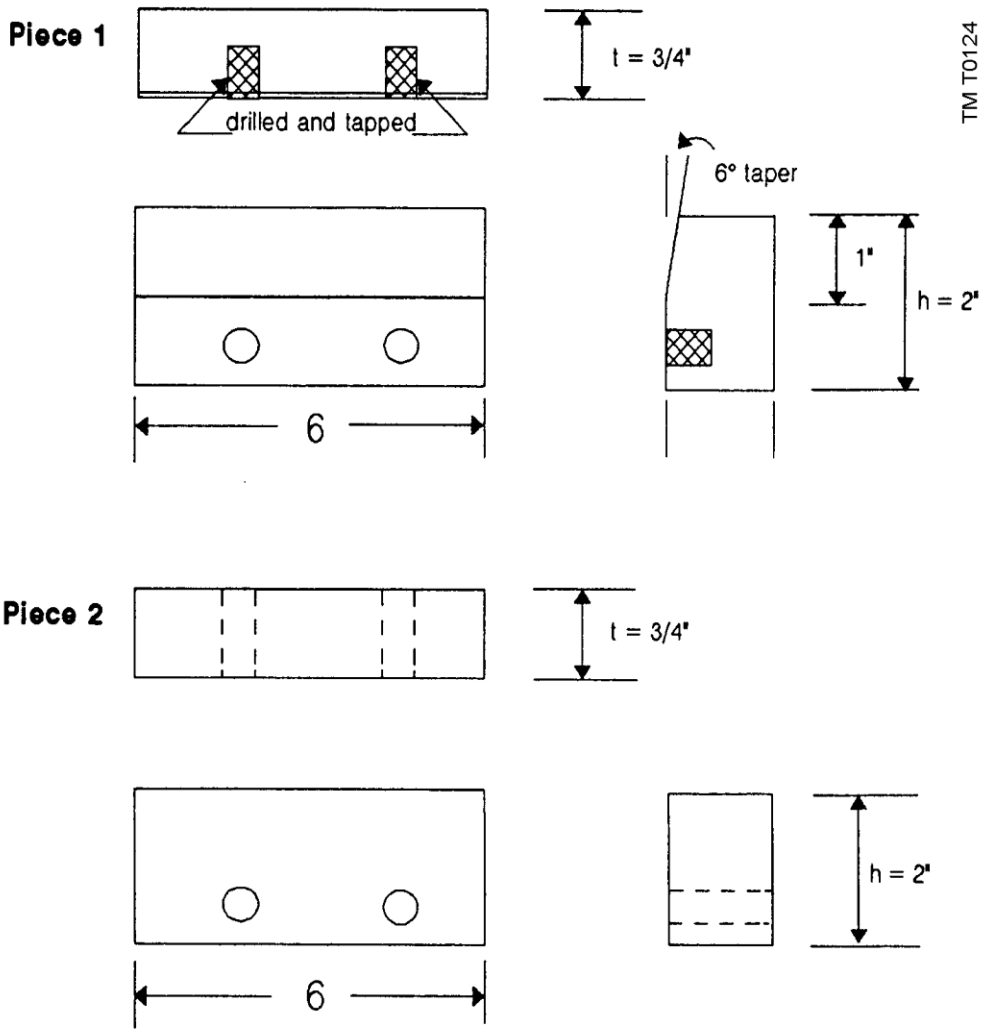


FIGURE D-1. Crevice fixture.

NOTES:

1. Surface finish shall be 250 Ra or better.

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 20GP-2024-001)

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